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Case Study / Case Report

Scrub Typhus Presenting as Severe Pneumonia and Acute Febrile Illness in a Malnourished Infant with Infantile Tremor Syndrome

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Abstract

Background: Scrub typhus is an important cause of acute undifferentiated febrile illness in endemic regions of India. Clinical manifestations are often nonspecific and may mimic pneumonia, meningitis, Kawasaki disease, or sepsis, making early diagnosis difficult. **Clinical Description:** A 6-month-old male infant presented with fever for 10 days, cough and respiratory distress for 4 days, and abdominal distension. The child had severe wasting and was underweight, with features suggestive of infantile tremor syndrome, including hyperpigmentation of the knuckles and tongue fasciculations. Examination revealed tachypnea, bilateral conjunctival congestion, cervical lymphadenopathy, hepatomegaly, and hypoxemia. Investigations showed anemia, thrombocytopenia, transaminitis, hyponatremia, low vitamin B12 levels, and left middle zone opacity on chest radiograph. Ultrasonography demonstrated ascites. Scrub typhus IgM was reactive while dengue serology was negative. **Management and Outcome:** The child was initially treated with empirical intravenous antibiotics; doxycycline and parenteral vitamin B12 were subsequently started after serological confirmation. Fever subsided within 48 hours with marked clinical improvement. **Conclusion:** Scrub typhus should be considered in infants presenting with prolonged fever, respiratory symptoms, hepatomegaly, thrombocytopenia, and transaminitis in endemic areas. Early diagnosis and prompt doxycycline therapy can be lifesaving. Coexisting malnutrition and infantile tremor syndrome may further obscure the diagnosis.

Key Words: Infantile tremor syndrome, Pediatric infectious disease, Prolonged fever, Scrub typhus, Severe pneumonia

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I. Introduction

Scrub typhus is a zoonotic infection caused by Orientiatsutsugamushi and transmitted through the bite of infected larval trombiculid mites (chiggers). It is increasingly recognized as a major cause of acute undifferentiated febrile illness in the "tsutsugamushi triangle," including several regions of India. Clinical manifestations range from mild febrile illness to severe multiorgan dysfunction involving the respiratory, neurological, hepatic, and cardiovascular systems.

In children, diagnosis is particularly challenging because early symptoms overlap with common tropical infections such as dengue, malaria, enteric fever, viral illnesses, and bacterial pneumonia. Presence of malnutrition or associated nutritional deficiencies may further complicate the clinical picture.

We report a case of scrub typhus in a 6-month-old infant presenting predominantly with severe pneumonia and features of infantile tremor syndrome, highlighting the diagnostic dilemma and importance of early suspicion in endemic regions.

II. Clinical Description

A 6-month-old male infant was admitted with fever for 10 days, cough and shortness of breath for 4 days, and abdominal distension for 4 days.

The child had a history of two previous hospitalizations for cough and respiratory distress, during which he received intravenous antibiotics for 5-7 days. He was born at term by normal vaginal delivery with a birth weight of 2.8 kg and was immunized appropriately for age. Feeding history revealed breastfeeding along with diluted cow's milk.

Family history was significant for smoking exposure at home and one neonatal death.

On examination, the infant appeared irritable and toxic-looking with severe wasting and underweight. He had tachypnea, cracked lips, bilateral conjunctival congestion, hyperpigmentation of knuckles, and tongue fasciculations. Oxygen saturation was 92% on room air. Bilateral cervical lymph nodes were palpable. Respiratory examination showed increased work of breathing with bilateral vesicular breath sounds and crepitations over the left hemithorax. Abdomen examination revealed hepatomegaly with liver palpable 2.5 cm below the right costal margin. Pallor was present, while icterus and clubbing were absent.

Initial differential diagnoses included severe pneumonia, Kawasaki disease, primary immunodeficiency, scrub typhus, and meningitis.

Management and Outcome:Laboratory investigations revealed hemoglobin 8.3 g/dL, total leukocyte count 18,000/cu mm with lymphocytic predominance, platelet count 1.2 lakh/cu mm, elevated transaminases (SGOT/SGPT: 82/46 IU/L), hyponatremia (Na+ 127 mEq/L), and low serum vitamin B12 levels. Chest radiograph demonstrated left middle zone opacity suggestive of pneumonia. Ultrasonography of abdomen revealed free fluid in the pelvis and hepatorenal pouch with contracted gall bladder. Dengue IgM was nonreactive, whereas scrub typhus IgM was reactive. Cerebrospinal fluid examination was unremarkable.

The child was initially managed with oxygen supplementation, nebulization, and empirical intravenous ceftriaxone and amikacin. Following confirmation of scrub typhus serology, doxycycline therapy was initiated along with parenteral vitamin B12 supplementation.

Clinical improvement was remarkable. Fever subsided within 48 hours of starting doxycycline, abdominal distension gradually decreased, respiratory distress improved, and the child became playful with reduced irritability.

The final diagnosis was scrub typhus with infantile tremor syndrome.

III. Discussion

Scrub typhus has emerged as a significant re-emerging pediatric infection in India. The disease often presents with nonspecific symptoms, resulting in delayed diagnosis and treatment. Classical eschar may be absent in many pediatric patients, particularly infants, thereby increasing diagnostic difficulty.

Our patient presented predominantly with prolonged fever, respiratory distress, hepatomegaly, thrombocytopenia, transaminitis, and hyponatremia, all of which are recognized manifestations of scrub typhus. Pulmonary involvement in scrub typhus may range from mild pneumonitis to severe pneumonia and acute respiratory distress syndrome.

The coexistence of severe malnutrition and infantile tremor syndrome further complicated the presentation. Features such as hyperpigmentation of knuckles, tongue fasciculations, macrocytosis, and low vitamin B12 levels supported the diagnosis of infantile tremor syndrome. Nutritional deficiencies may impair immunity and predispose to severe infections.

An important clinical clue in this case was the dramatic defervescence within 48 hours of initiating doxycycline, which strongly supported the diagnosis of scrub typhus. Early initiation of appropriate therapy significantly reduces morbidity and mortality.

This case emphasizes that scrub typhus should remain an important differential diagnosis in children presenting with prolonged fever and respiratory symptoms in endemic areas, especially when associated with thrombocytopenia, transaminitis, or hepatomegaly.

Learning Points

- Scrub typhus should be considered in prolonged febrile illness with respiratory distress in endemic regions.
- Thrombocytopenia, transaminitis, hyponatremia, and hepatomegaly are important diagnostic clues.
- Absence of eschar does not exclude scrub typhus.
- Early doxycycline therapy produces rapid clinical improvement.
- Coexisting malnutrition and vitamin B12 deficiency may complicate presentation and management.

Consent

Written informed consent was obtained from the child's guardians for publication of this case report and clinical details.

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